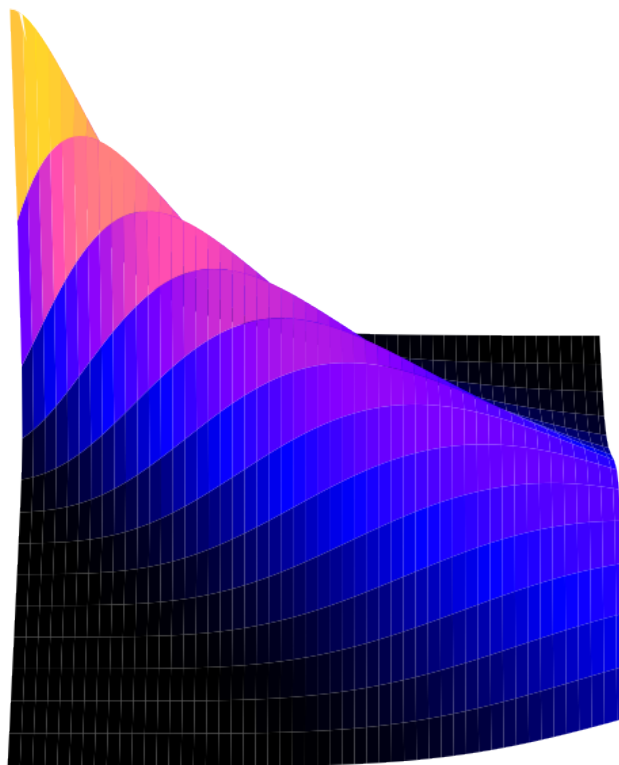


Poisson Models



1 Introduction

Poisson distribution, introduced by [Siméon Denis Poisson](#) in 1837, describes the incidence of events during a fixed time interval. Originally developed to model rare, discrete events, this distribution has found widespread applications across diverse fields ranging from estimating the number of customer service calls to modeling insurance claims, network traffic accidents, and other rare events.

While this distribution rarely fits real-world data perfectly as it relies on some very strong assumptions, it presents an interesting opportunity to model certain processes that fit within the Poisson framework.

Poisson-distributed data can be modeled using parametric linear models, Bayesian inference, or machine learning algorithms such as gradient-boosted trees with a Poisson objective. This text focuses on the linear and ML models and gives several examples of how to apply them in Python. We also provide a Python implementation of the Poisson generalized linear model (GLM) using the Fisher scoring algorithm.

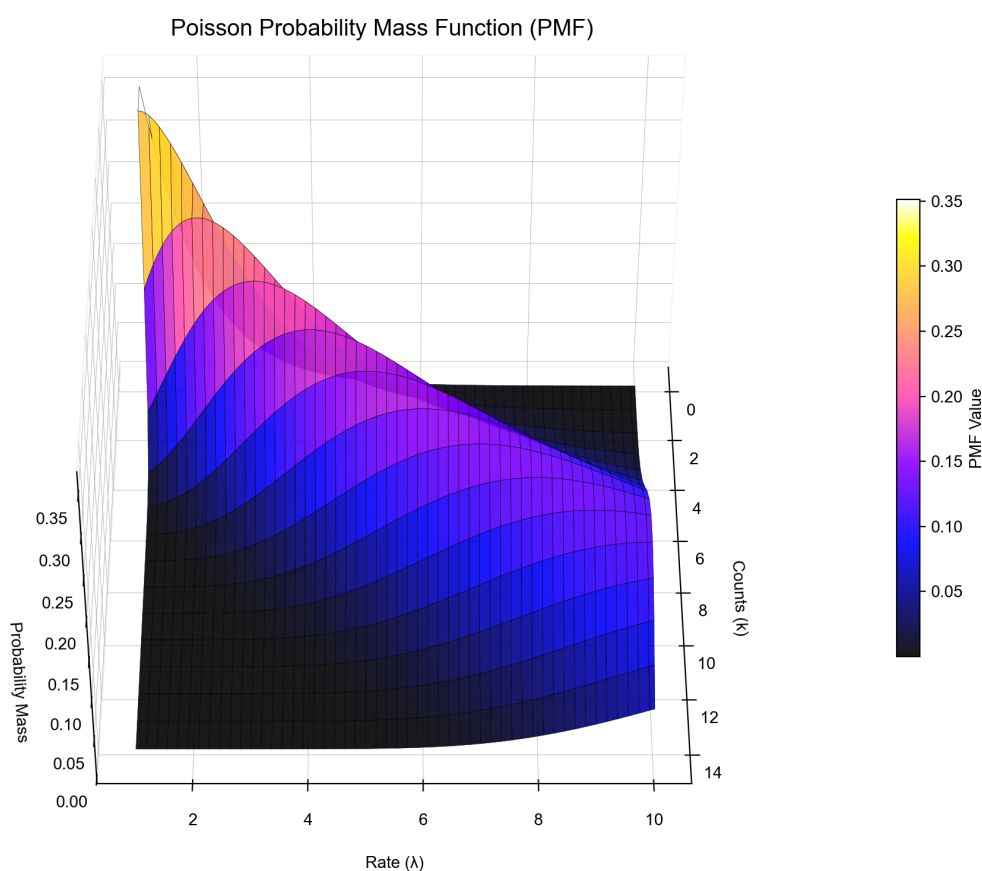


Figure 1: 3D visualization of the Poisson Probability Mass Function

2 Poisson Distribution

The Poisson distribution models the probability of a given number of events occurring in a fixed interval of time or space, assuming these events occur independently at a constant rate.

The Poisson probability mass function (PMF) that describes the probability of incidents is:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!} \quad (1)$$

Where:

- k is the number of occurrences
- λ is the expected number of occurrences (rate parameter)

Key Properties:

- The mean and variance of the Poisson distribution are both equal to λ .
- Suitable for modeling rare events over a fixed period or space.
- As λ increases, the Poisson distribution approaches a normal distribution.
- The probability of zero occurrences ($P(X = 0) = e^{-\lambda}$) is particularly important in many applications.

Poisson distribution focuses on the incidence rate of rare event counts in isolation from total possible events, unlike proportions (risk or prevalence) modeling commonly performed with logistic regression that uses Bernoulli likelihood that does require the total number of events.

When exponentiated, Poisson regression coefficients may be interpreted as incidence rate ratios. The output of the model is a distribution of $E[Y|X]$ (expected value of Y given X) that is Poisson distributed, so for a given observation i , the prediction of Poisson regression does not represent a predicted count, but an expected mean number of events for a given set of input features.¹

Given that variance and mean equality is a very strong assumption, Poisson models often fail to fit the real-world data, but there are examples when they do.

The 3D visualization above represents the Poisson Probability Mass Function (PMF) with:

- λ (**Rate**) on the horizontal axis: the expected value (mean) of the distribution.
- k (**Counts**) on the vertical axis: the number of events observed.
- **PMF value** on the z-axis: the probability of observing exactly k events given a Poisson process with rate λ .

The chart shows how probability changes across different values of λ and k . For small λ values (1-2), probabilities peak at lower k values (0-1). As λ increases (toward 10), the probability mass shifts right, with peaks occurring at higher k values. The color gradient from dark blue to yellow represents increasing probability values.

3 Horse Kick Data: Classic Rare Events Example

Deaths from horse kicks might seem like extraordinarily rare events, but for those serving in the Prussian cavalry during the 19th century it was not so uncommon. What makes the case of Prussian horse kicks particularly valuable to statisticians like Sir Ronald Aylmer Fisher is how it can almost perfectly illustrate the Poisson distribution.

In *Statistical Methods for Research Workers* (1925), Fisher used Ladislaus Bortkewitch's Prussian Horse Kicks dataset to introduce Poisson series. The dataset used by Fisher contains incidence of horse-kick accidents for twenty successive years spanning between 1875 and 1894. Notably, both L. Bortkewitsch and R.A. Fisher used only a subset of data (200 records out of 280), since corps denoted G, I, VI, and XI deviated significantly from the average.

What Fisher notes is that, with its single parameter $\lambda = 0.61$ (the average death rate), the Poisson distribution predicted almost exactly these same numbers. The distribution expected 108.7 instances of zero deaths, 66.3 instances of one death, and so on.

¹In many applications such as epidemiology, an offset term is included to account for exposure time or space (e.g., forest density, population size). This adjusts the expected response by the exposure, and the model is fit with an additional constant that presents the results as an incidence rate rather than an absolute count.

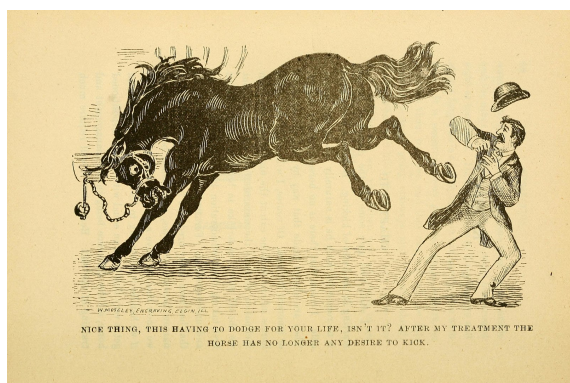


Figure 2: A practical system of colt training

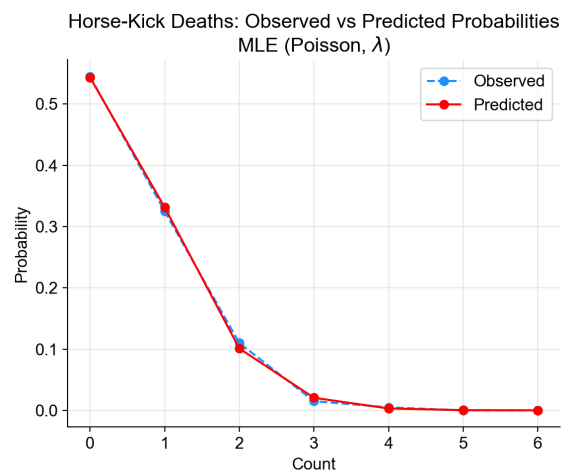


Figure 3: Horse-Kick Deaths: Observed vs Predicted Probabilities

Count	Frequency Observed	Frequency Predicted
0	109	108.67
1	65	66.29
2	22	20.22
3	3	4.11
4	1	0.63
5	0	0.08
6	0	0.01

Table 1: Observed vs. Predicted Frequencies for Horse Kick Deaths

Poisson Series Example

Poisson series example from R. Fisher (1925, 14th edition, p. 55), based on 200 observations from Bortkewitch's data (Prussian Horse Kick Data).

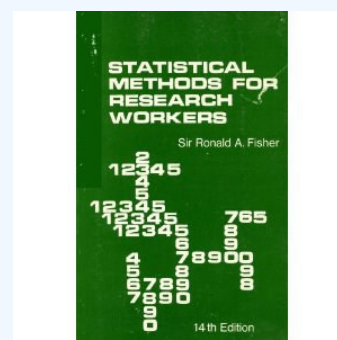
It may be shown theoretically that if the probability of an event is exceedingly small, but a sufficiently large number of independent cases are taken to obtain a number of occurrences, then this number will be distributed in the Poisson series.

For example, the chance of a man being killed by horsekick on any one day is exceedingly small, but if an army corps of men are exposed to this risk for a year, often one or more of them will be killed in this way.

The following data (Bortkewitch's data) were obtained from the records of ten army corps for twenty years, supplying 200 such observations. The average, $\bar{x}$, is 0.61, and taking this as an estimate of m , the numbers calculated agree excellently with those observed.

Data sources:

- McElreath 2020. Statistical Rethinking, 2nd edition, CRC Press
- VonBort



The Poisson distribution provides an excellent fit for this rare event data. This confirms the theoretical expectation that when an event probability is extremely small but observed over many cases, the occurrences follow a Poisson distribution. However, this is rarely the case in real-world data.

4 German Doctor Visits

In Germany in the 1980s, health researchers collected data on how often people visited their doctors for the German health reform. The response variable is the number of visits, a significant share of which are zero.

Dataset: rwm ("COUNT" R Package)

This dataset uses the German health reform for the years 1984-1988, providing health information for years prior to the health reform.

- **Type:** Health dataset
- **Total Observations:** 27,326
- **Key Features:**
 - **docvis:** Number of visits to the doctor during the year (0-121)
 - **age:** Age range of individuals (25-64)
 - **educ:** Years of formal education (7-18)
 - **hhninc:** Household yearly income in DM/1000

Subset: rwm1984 ("COUNT" R Package)

This subset focuses on the German health registry for the year 1984. It contains 3,874 observations with the following variables:

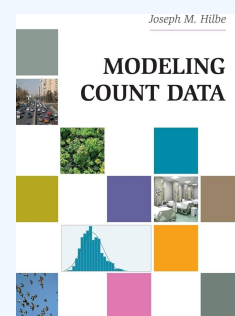
- **docvis:** Number of visits to the doctor (0-121)
- **hospvis:** Number of days in hospital (0-51)
- **edlevel:** Educational level (categorical: 1-4)
- **age:** Age range of individuals (25-64)
- **outwork:** Indicator for being out of work (1=out of work, 0=working)
- **female:** Indicator for gender (1=female, 0=male)
- **married:** Indicator for marital status (1=married, 0=not married)
- **kids:** Indicator for having children (1=yes, 0=no)
- **hhninc:** Household yearly income in DM/1000
- **educ:** Years of formal education (7-18)
- **self:** Indicator for self-employment (1=self-employed, 0=not)
- **edlevel1:** Not a high school graduate (1/0)
- **edlevel2:** High school graduate (1/0)
- **edlevel3:** University/college graduate (1/0)
- **edlevel4:** Graduate school (1/0)

German Health Registry Dataset Information

Hilbe, Joseph, M (2014), *Modeling Count Data*, Cambridge University Press
Hilbe, Joseph M (2011), *Negative Binomial Regression*, Cambridge University Press
Hilbe, J. and W. Greene (2008), *Count Response Regression Models*, in ed. C.R. Rao, J.P. Miller, and D.C. Rao, *Epidemiology and Medical Statistics, Elsevier Handbook of Statistics Series*. London, UK: Elsevier.

Model from Modeling Count Data by Joseph M. Hilbe

For example, given a random sample of 1000 patient observations from the German Health Survey for the year 1984, the following Poisson model output explains the years expected number of doctor visits on the basis of gender and marital status, both recorded as binary (1/0) variables, and the continuous predictor, age.



The German health registry sample included 3,874 individuals and recorded:

- 43% of people (1,611) had zero doctor visits
- 23% had one or two visits
- 18% had three to five visits
- 16% had six or more visits

Unlike the horse kick data, this everyday counting scenario proved troublesome for the standard Poisson model, which predicted:

- Only 7.6% of people would have zero visits (severely underpredicting)
- 36.6% would have one or two visits (overpredicting)
- 43% would have three to five visits (greatly overpredicting)
- 12.8% would have six or more visits (underpredicting)

4.1 Why Excess Zeros Are a Problem

When the number of observed zeros exceeds what the Poisson distribution predicts, we have **zero inflation**. This creates problems because:

- **Variance > Mean** (violates Poisson assumptions)
- **Overdispersion** inflates standard errors and reduces model accuracy
- May reflect a **two-process model**: some units never experience events (structural zeros)

A commonly accepted alternative is to use Negative Binomial model for this type of problems. The Negative Binomial distribution adds an extra parameter (dispersion parameter) that allows variance to exceed the mean, making it much more flexible for overdispersed data. This explains why it performs so much better than the standard Poisson model for the doctor visits data.

Negative Binomial Model

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Another alternative is to use ML algorithms such as gradient-boosted decision trees to model Poisson process.

Below we show fit metrics for different models:

Metric	Poisson Model	Negative Binomial	XGBoost	CatBoost
Log-Likelihood	-15,502.51	-8,653.40	-4,465.58	-8,530.77

Table 2: Model Comparison by Log-Likelihood

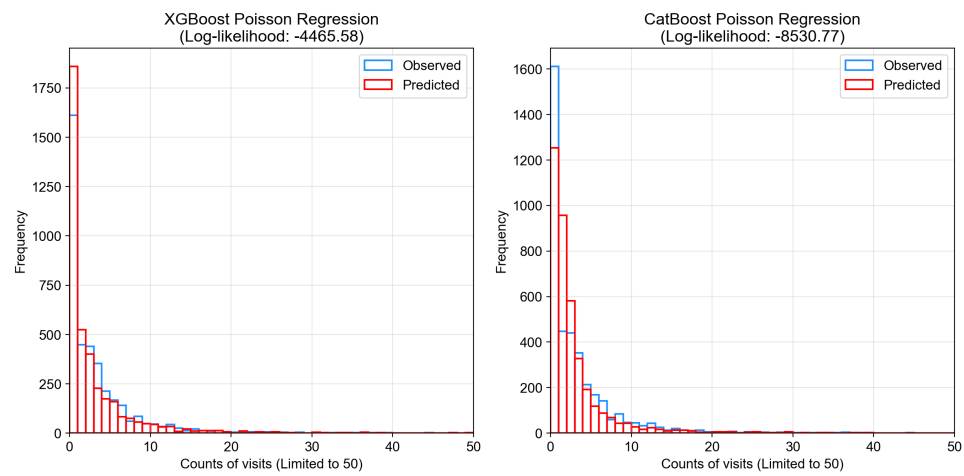


Figure 5: Histogram Comparison

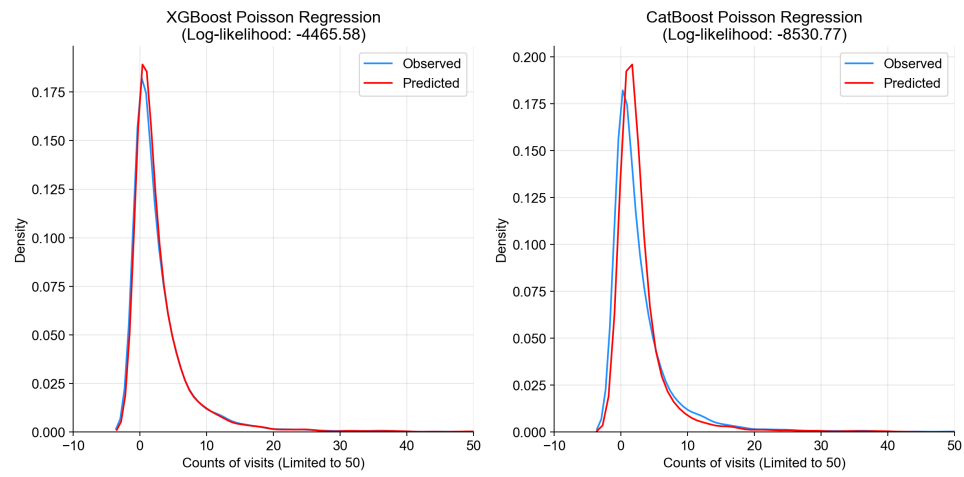


Figure 6: Density Comparison

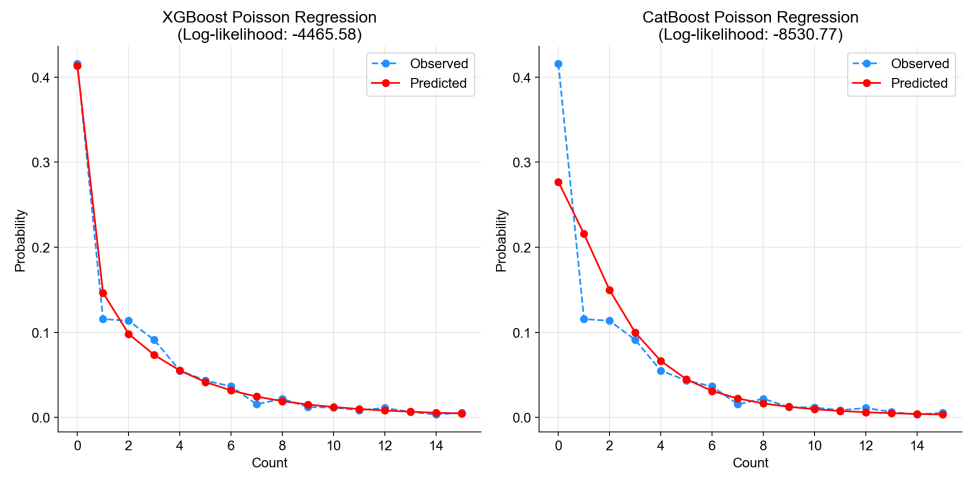


Figure 7: Observed vs Predicted Probabilities

The standard Poisson model is clearly the poorest fit, while the Negative Binomial model provides a better alternative. Gradient boosting methods show superior predictive performance partially due to intentional overfitting for this example. The goal is to show that ML algorithms can capture nonlinear interactions and thus provide a better fit for the data in certain cases.

5 Implementing Maximum Likelihood Poisson Regression

When fitting Poisson regression models with Maximum Likelihood Estimation (MLE), the process is very similar to logistic regression with one key difference being that the information matrix is not based on Bernoulli variance, $p(1 - p)$, but on the expected count parameter μ :

```
# Fisher Scoring Algorithm for Poisson Regression
# This iterative algorithm finds the Maximum Likelihood Estimates (MLE)
# for the parameters of a Poisson regression model

for iteration in range(max_iterations):
    # Step 1: Calculate the linear predictor and expected values
    eta = X @ weights          # Linear predictor (Xβ*)
    mu = np.exp(eta)           # Transform to mean using link function

    # Step 2: Calculate the score vector (gradient of log-likelihood)
    score = X.T @ (y - mu)     # X'(y - μ)

    # Step 3: Calculate the Fisher information matrix
    W = np.diag(mu)            # Weight matrix with μ on diagonal
    information = X.T @ W @ X   # X'WX

    # Step 4: Update parameters using Fisher scoring
    # Solve the equation: information * delta = score
    delta = np.linalg.solve(information, score)
    beta_new = weights + delta
```

We can use `fisher-scoring` package to fit a Poisson model and visualize the results:

```
# !pip install fisher-scoring==2.0.4
from fisher_scoring import PoissonRegression
from fisher_scoring.utils import plot_observed_vs_predicted

# Create a class instance
poisson_model = PoissonRegression(use_bias=False)

# Fit the Poisson regression model
poisson_model.fit(X, y)

# Make predictions
mu = poisson_model.predict(X)

# Plot Observed vs Predicted probabilities
df_results = plot_observed_vs_predicted(
    y,
    mu,
    max_count=15,
    title="Doctor Visits Data: Observed vs Predicted Probabilities",
    model_name="Poisson Regression",
)
```


Fisher Scoring Package

The Fisher scoring algorithm is an optimization technique used for finding maximum likelihood estimates. You can install the package with:

```
pip install fisher-scoring==2.0.4
```

The Fisher scoring algorithm is an optimization technique used for finding maximum likelihood estimates.

6 Conclusion

Poisson distributions provide a powerful framework for modeling count data, especially for rare events and random point processes. Understanding when they apply—and recognizing their limitations with overdispersed data, excess zeros, and raw totals—enables more accurate analysis of real-world count phenomena.

The examples we've explored demonstrate both the strengths and limitations of Poisson models:

1. The Horse Kick data shows how well Poisson distributions can model truly rare events
2. The German Doctor Visits data reveals the challenges of overdispersion and zero-inflation
3. Modern approaches like Negative Binomial models and gradient boosting can overcome these limitations

When working with count data, it's important to:

- Test for overdispersion
- Consider zero-inflation
- Compare multiple modeling approaches

Report created by [Denis Burakov](#)

Additional Resources

- [GitHub Repository](#)
- [Notebook on NBViewer](#)
- [fisher-scoring Python package](#)



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